One Bad Apple: How Does the Success of a Fungicide Resistance Management Strategy Respond to a Proportion of Growers Who Do Not Follow Recommendations?

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Introduction

- We modelled how the heterogenous usage of fungicide to treat Septoria tritici blotch (STB) of winter wheat affects the spatial spread and selection of fungicide resistance over multiple growing seasons in the causative pathogen *Zymoseptoria tritici* (*Zt*) population.
- The major source of primary inoculum is widespread airborne *Zt* ascospores [1]. Hence local heterogeneity in fungicide usage impacts local fungicide resistance selection over multiple growing seasons when compared to fields treated in isolation.
- An established epidemiological model [2] for the selection of qualitative resistance has been extended to a spatially implicit model and a separate spatially explicit model. Growers are split into two groups corresponding to the use of high or low fungicide concentration programmes.

Results

Within-Season Model, Isolated Fungicide Programme

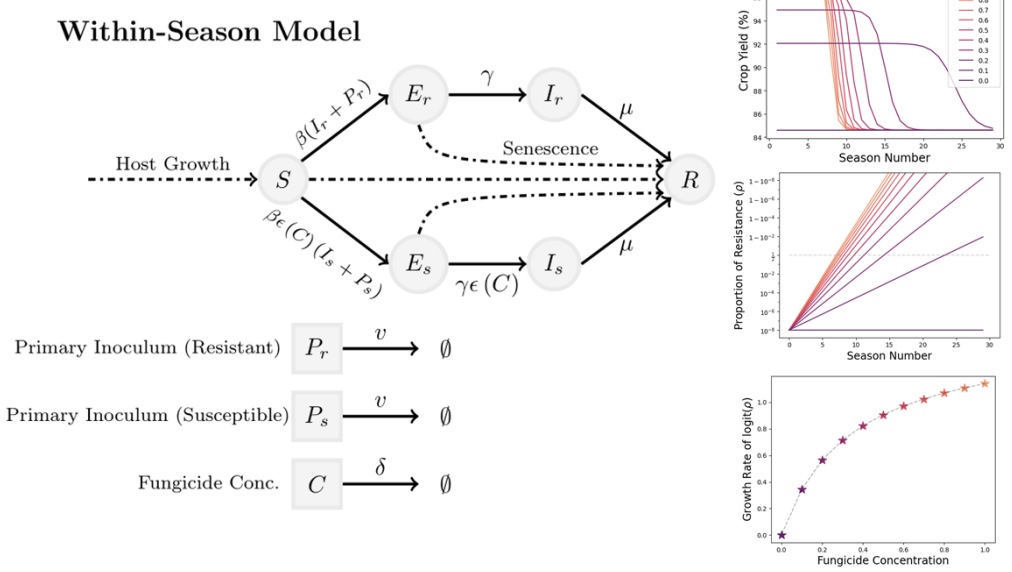


Figure 1. (Left) Schematic of the within-season compartmental model. Subscript *r* and *s* correspond to disease caused by fungicide resistant or susceptible genotype *Zt*. (Right) Plots showing Crop Yield (%) and Proportion of Fungicide Resistant Genotype *Zt* (ρ) against Season Number and Growth Rate of ρ against Fungicide Concentration.

- The proportion of resistant genotype *Zt* grows logistically over multiple seasons; the rate of logistic growth depends on the concentration of fungicide applied.

Spatially Implicit Model

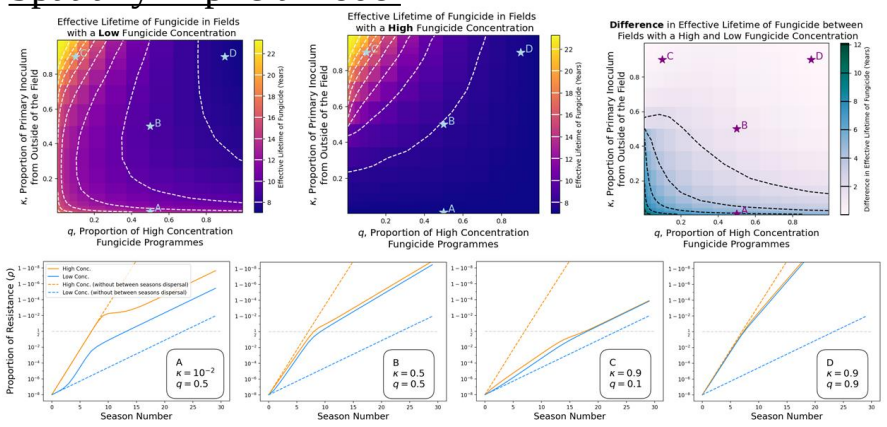


Figure 2. (Top) Effective lifetime of Fungicide (years) until fungicide resistant genotype *Zt* is the dominant strain (ρ = 0.5) over 2D parameter space; the proportion of primary inoculum from outside of the field (κ) and the proportion of high concentration fungicide programmes (q). (Bottom) Cases A, B, C, D show the selection of fungicide resistance over time for given parameter values.

- Two necessary conditions for the rate of fungicide resistance selection to be reduced through decreased fungicide usage are: (i) most growers use low concentration fungicide programmes; (ii) there is high dispersion of primary inoculum between fields.

Spatially Explicit Model

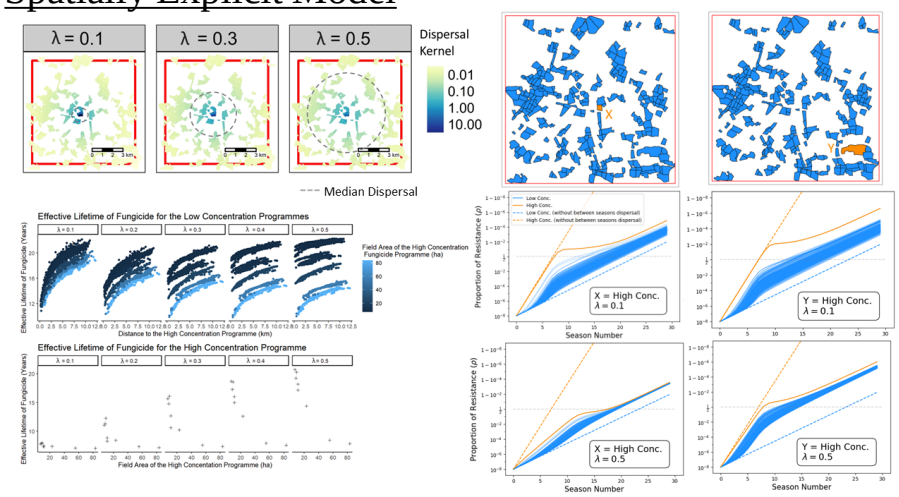


Figure 3. (Top left) Inverse power law dispersal kernel $= (d + \lambda)^{-\alpha}$ over the host landscape, varying λ with fixed $\alpha = -2.5$. (Right) Selection of resistance with high (orange) and low (blue) concentration fungicide programmes for cases X and Y. (Bottom left) Combined results for 8 simulations with randomly chosen field of high concentration showing the effect of Field Area (ha) on Effective Lifetime of Fungicide (years).

- Ascospore dispersal has been modelled explicitly using an inverse power law dispersal kernel [3]. Spatially explicit modelling shows that condition (i) and (ii) are not sufficient for localities with a grower using a high concentration programme in field of large area.

Conclusion

- Area wide management and grower cooperation is needed to optimise the management of fungicide resistance.
- Future work includes empirical validation of the models developed including an assessment of predictive power.

References: [1] Shaw, M. W., & Royle, D. J. (1989, March). Airborne inoculum as a major source of Septoria tritici (Mycosphaerella graminicola) infections in winter wheat crops in the UK. Plant Pathology, 38 (1), 35–43. [2] Hobbelen, P. H. F., Paveley, N. D., Fraaije, B. A., Lucas, J. A., & van den Bosch, F. (2011, April). Derivation and testing of a model to predict selection for fungicide resistance: Testing a fungicide resistance model. Plant Pathology, 60 (2), 304–313. [3] Karisto, P., Suffert, F., & Mikaberidze, A. (2023). Spatially explicit ecological modelling improves empirical characterization of plant pathogen dispersal. Plant-EnvironmentInteractions, 4, 86– 96.